Muhammed Can Küçükaslan

Senior Backend Software Engineer

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Senior backend engineer building and operating high-traffic Python services, routing systems, and reliable cloud infrastructure. Also builds in Go, Java, and C++, and contributes to open-source tools.

Work Experience

Aug. 2026 – present
Getir
Senior Backend Engineer I

- Diagnosed uneven pod traffic caused by persistent HTTP connections and Kubernetes ClusterIP's connection-level load balancing. Introduced L7 request routing; client timeouts fell by **more than 50%**, and requests completed under **20 ms rose from 70% to 84%**.
- Traced oversized Kubernetes resource requests and limits to reclaimable filesystem cache. Right-sized deployments; relevant EC2 spend fell by **$30/day**, roughly **$11k annualized**.
- Standardized production logging and investigated distributed trace propagation across New Relic and OpenTelemetry services.

Feb. 2025 – July 2026
Getir
Backend Engineer II

- Owned production **Python/FastAPI** routing and duration-estimation services, including services handling **350,000+ requests/minute at peak**.
- Collaborated on migrating an internal vehicle routing service from Python to C++, **doubling performance** and saving **$10,000+ in monthly recurring costs**.
- Customized OSRM deployments using historical delivery data and GPS traces, with Airflow, PostgreSQL, and Redis.
- Audited Kubernetes resource limits and HPA policies, reducing annual infrastructure costs by **$20,000**.
- Deployed Kafka, MongoDB, Kubernetes, and EC2 infrastructure with strict ACLs using Terraform and Cloud CDKs; migrated CI/CD from Bitbucket to **GitHub Actions**.
- Led vehicle-tracking IoT integration with device-specific protocols in **Java and Go** microservices.
- Fixed recurring service memory exhaustion during RabbitMQ maintenance by changing a shared logging library to spill buffered records to disk under memory pressure.
- Mentored interns and junior engineers and wrote company-wide MongoDB upgrade guides.

Jan. 2024 – Jan. 2025
Getir
Backend Engineer I

- Built and maintained **Go and Java/Spring** services, including a courier rewards service using Kafka and Flink; used ShedLock for distributed scheduling.
- Improved MongoDB query performance and helped reduce database costs by **more than $2,000/month**.

July – Nov. 2023 **OTO Global · Remote**
Backend Engineering Intern and Part-time Engineer

- Developed Java/Spring services on Google Cloud Run and Compute Engine; diagnosed production issues with metrics and logs.

June – Sep. 2022 **Peak Games**
Big Data Platform Engineering Intern

- Contributed to **s5cmd**, a Go CLI for S3 operations. Added S3 versioning, shell completion, and reliability fixes; redesigned large-scale sync using external sorting.

Selected Open Source and Writing

- **Argo Rollouts (Go, Kubernetes):** Investigated production retry failures and submitted a controller fix for stale analysis state across Blue/Green and Canary rollouts. Preserves readiness and promotion gates, with regression coverage for retries and transient API failures. PR #5081 — open.
- **s5cmd (Go, AWS S3):** Reworked sync with external sorting, reducing memory from **~10 GB to ~1.5 GB for one million files**, at a cost of ~20% slower runtime in the published benchmark.
- **osmium-tool (C++):** Diagnosed a long-standing geographic extraction bug, produced a focused reproduction, and contributed the upstream fix with regression coverage. Investigation.
- **Kubernetes request routing:** Wrote about connection-level load balancing and built an interactive HTTP routing simulator.

Skills

Languages: Python, Go, Java, C++, SQL.
Backend and data: FastAPI, Uvicorn, Spring, Kafka, Redis, MongoDB, PostgreSQL, OSRM.
Infrastructure: Kubernetes, Docker, AWS (EC2, S3), Terraform, GitHub Actions, Google Cloud.
Observability: New Relic, OpenTelemetry, distributed tracing, structured logging, performance analysis.

Education

Bilkent University — B.S. in **Computer Engineering** and B.S. in **Mathematics** (double major), January 2024. **Magna cum laude**; GPAs **3.66/4.00** and **3.65/4.00**. Comprehensive Scholarship (100% tuition waiver).